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Cyclic Liquid Nitrogen Fracturing: A New Reservoir Stimulation Method to Enhance the Hot Dry Rock Fracturing Performance

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Abstract

Geothermal energy offers a promising solution to address the challenges posed by global warming and the carbon crisis. Fractures serve as primary conduits for fluid flow and heat transfer in geothermal reservoirs such as Hot dry rock (HDR) reservoir, thereby exerting a significant influence on the performance of heat extraction. Traditional hydraulic fracturing in hot dry rock (HDR) reservoirs often leads to excessively high breakdown pressures, generates primarily a single dominant fracture, and may elevate the probability of induced seismicity. To overcome these limitations, this research investigates an alternative stimulation approach termed cyclic liquid nitrogen (LN₂) fracturing, which integrates the concepts of cyclic soft stimulation (CSS) and cryogenic LN₂ fracturing. In this technique, liquid nitrogen at ultra-low temperatures is periodically injected with alternating phases of high-rate pumping and low-rate (or suspended) injection. Under such cyclic loading, the reservoir formation experiences repeated thermal shocks combined with fluid pressure fluctuations, resulting in progressive fatigue damage of the rock. This process facilitates fracture nucleation, enhances fracture growth and branching, and ultimately promotes the generation of a more intricate fracture network, thereby enlarging the effective stimulated reservoir volume. To verify the feasibility of cyclic LN₂ fracturing on HDR, a thermo-hydro-mechanical-damage (THMD) model for cyclic LN₂ fracturing on HDR was established considering the effect of LN₂ physical properties and rock heterogeneity based on the Phase-field cohesive zone method (PF-CZM). The fracturing performance of three cyclic LN₂ injection schemes, namely balanced injection, pulsating injection and stepped injection scheme, were investigated. The results demonstrate that cyclic LN₂ injection further enhances LN₂ fracturing performance. In the pulsating injection scheme, the damage ratio increased by 120.1% when the cycle number increased from 1 to 3. Pulsating injection showed better results compared to balanced injection when the LN₂ injection time is the same. However, the improvement in fracturing performance became negligible after 3 cycles. In cyclic LN₂ fracturing of HDR reservoirs, the coordinated control of thermal stress and fluid pressure plays a decisive role in inducing fracture bifurcation and promoting the evolution of a complex fracture system. In the stepped injection scheme, numerous thermally-induced fractures are easily formed

around the wellbore but the main fracture extend shorter when the pressure increased amplitude is smaller. Conversely, with a larger pressure increased amplitude, the expectation is for the main fracture to expand far away from the direction of maximum horizontal stress. These observations provide important theoretical insights, offering a foundation for improving the efficiency and ensuring the safety of HDR geothermal resource development.

Keywords: Enhanced Geothermal System, Hot dry rock, Cyclic liquid nitrogen fracturing, Thermal stress, Fatigue damage

Introduction

Geothermal energy is increasingly recognized as a sustainable pathway to mitigate the impacts of climate change and the ongoing carbon crisis. Achieving both commercial feasibility and long-term sustainability of geothermal development, however, requires the efficient and optimized exploitation of available geothermal resources. Fractures serve as primary conduits for fluid flow and heat transfer in geothermal reservoirs such as Hot dry rock (HDR) reservoir, thereby exerting a significant influence on the performance of heat extraction. In many cases, hydraulic fracturing techniques are employed to augment natural fractures by creating artificial pathways that connect injection and production wells. This integration of natural and induced fractures optimizes the overall efficiency of the geothermal system (Paulillo et al. 2020, Zhong et al. 2023). However, some key challenges are met in HDR stimulation by using conventional fracturing method, especially in generally forming 2D planar fracture patterns and requiring high breakdown pressure (Watanabe, Egawa, et al. 2017, Watanabe, Numakura, et al. 2017, Park et al. 2017). 2D planar fracture patterns may lead to the thermal breakthrough and cannot extract enough heat energy compared with complex (branched) fracture networks (Yin et al. 2021, McClure and Horne 2014). Higher breakdown pressure tends to bring higher operating costs and safety risks such as seismic event in filed applications (Park et al. 2017).

An alternative stimulation strategy, referred to as fatigue hydraulic fracturing (FHF) (Zang et al. 2013) or cyclic soft stimulation (CCS) (Hofmann et al. 2018), introduces novel injection protocols to overcome the challenges of conventional fracturing in HDR reservoirs. The central concept of FHF/CCS is to apply cyclic fluid injection with alternating high- and low-rate stages while maintaining relatively low pressures and limited net fluid volumes (Zang et al. 2018). A series of laboratory investigations and pilot-scale field applications have evaluated the effectiveness of cyclic fracturing with respect to breakdown pressure reduction, fracture propagation behavior, and the magnitude of induced seismic events (Zhuang et al. 2016, Zhuang et al. 2019, Zhuang and Zang 2021, Patel et al. 2017, Zang et al. 2013). Results consistently demonstrate that cyclic injection can mitigate seismic risks by regulating crack-tip stress release. Moreover, water-based cyclic stimulation has been observed to lower the required breakdown pressure and promote the development of more intricate fracture networks characterized by numerous branches and finer apertures, in contrast to conventional single-cycle hydraulic fracturing.

Recently, cryogenic fracturing utilizing liquid nitrogen (LN₂) as the fracturing fluid has gained significant attention in Enhanced Geothermal System (EGS) (Huang et al. 2020, Yang et al. 2019, Yang et al. 2021, Sun et al. 2022). LN₂, with its temperature of −196 °C at atmospheric pressure, creates a substantial thermal gradient when injected into the reservoir formation, leading to the formation of numerous fractures. Laboratory studies have shown the advantages of LN₂ fracturing, including reducing initiation/breakdown pressure, enhancing permeability around the borehole, and creating complex fracture networks within rocks compared to conventional hydraulic fracturing (Wu et al. 2021, Qu et al. 2022, Cha et al. 2018). Inspired by cyclic hydraulic fracturing, cyclic LN₂ fracturing has been proposed and validated to enhance the performance of LN₂ fracturing through cyclic injection. (Hong et al. 2023, Yang et al. 2023). However, current investigations mainly rely on laboratory testing. Rock samples are typically subjected to a closed

true triaxial-loading and heating system in a high-temperature and high-pressure environment, and fracture patterns are observed after the fracturing tests. Numerical simulation is a promising approach to overcome these limitations. However, the coupled thermal-hydraulic effects on the initiation and propagation behavior of induced fractures during cyclic LN₂ fracturing, as well as resulting fracture patterns, have been scarcely reported. Simulating hydro-thermally driven fracture evolution in porous media remains highly challenging, primarily because of the intricate coupling between fluid flow, heat transfer, and solid deformation, as well as the resultant development of complex fracture networks.

To address this research gap, a novel thermo-hydro-mechanical-damage (THMD) coupling framework that incorporates rock heterogeneity is developed using the phase-field cohesive zone method (PF-CZM). This approach enables a detailed investigation of fracture initiation, propagation dynamics, and the formation of complex fracture networks during cyclic LN₂ stimulation. Within this framework, the fracturing responses of three distinct LN₂ cyclic injection strategies—balanced, pulsating, and stepped modes—were systematically analyzed. The outcomes of this work are expected to offer both theoretical insights and numerical modeling support for the efficient development of HDR geothermal resources.

Methodology

Governing equations

Cyclic LN₂ fracturing in HDR reservoirs represents a multi-physics coupling process, involving the interaction of fluid flow, geomechanical response, heat transfer, and progressive rock damage. Within this framework, the migration of fracturing fluids throughout the computational domain is governed by Darcy's law, which can be expressed as (Zhou et al. 2018, Zhou and Zhuang 2020)

$$\rho_f S \frac{\partial P}{\partial t} - \nabla \cdot (\rho_f \mathbf{v}) = q_m - \rho_f \alpha_B \chi_r \frac{\partial \varepsilon_{vol}}{\partial t}, \quad (1)$$

where ρ_f [kg/m³] denotes the density of the fracturing fluid; α_B represents the Biot coefficient; $\mathbf{v}$ [m/s] is the Darcy's velocity; q_m [kg/(m³·s)] refers to the fluid source term; P [Pa] is the pore fluid pressure at the current state; χ_r is an indicator function and ε_{vol} denotes the volumetric strain. In particular, the storage coefficient S , which incorporates both fluid properties and the Biot coefficient, can be defined as

$$S = \phi c_f + \frac{(\alpha_B - \phi)(1 - \alpha_B)}{K_V}, \quad (2)$$

here ϕ denotes the porosity of the rock matrix; c_f [Pa⁻¹] is the compressibility of the fracturing fluid and K_V [MPa] represents the bulk modulus of the rock matrix.

The local thermal equilibrium (LTE) assumption is adopted to characterize heat transfer between the rock matrix and the fracturing fluid. Under this framework, the energy conservation equation can be formulated as

$$(\rho C_p)_{eff} \frac{\partial T}{\partial t} + \rho_f C_{p,f} \mathbf{v} \cdot \nabla T - \nabla \cdot (\lambda_{eff} \nabla T) = Q_T \quad (3)$$

where $(\rho C_p)_{eff}$ [J/(K·m³)] and λ_{eff} [W/(m·K)] denote the effective volumetric heat capacity and effective thermal conductivity, respectively; $C_{p,f}$ [J/(kg·K)] is the specific heat capacity of the fracturing fluid; T [K] represents the temperature at the current state; Q_T [W/m³] denotes the heat source term.

The mechanical equilibrium equation is adopted to describe the quasi-static problems, which can be written as

$$\nabla \cdot \boldsymbol{\sigma} + \mathbf{b} = 0, \quad (4)$$

here $\mathbf{b}$ represents the body force and $\boldsymbol{\sigma}$ is the Cauchy stress tensor, which can be expressed as

$$\boldsymbol{\sigma} = \sigma_{eff} \mathbf{I} - \alpha_B (P - \bar{P}) \mathbf{I}, \quad (5)$$

where σ_{eff} is the effective stress tensor; $\bar{P}$ [Pa] is the fluid pressure; I represents the identity tensor. The term $\alpha_B(P - \bar{P})I$ denotes the contribution of fluid pressure to the Cauchy stress. Besides, the effective stress tensor σ_{eff} denotes the average stress. When accounting for thermal expansion, the stress tensor can be expressed as (Chen and Wu 2022)

$$\sigma_{eff} = \mathbf{C} : \varepsilon_m = \mathbf{C} : (\varepsilon - \varepsilon_t), \quad (6)$$

where $\mathbf{C}$ represents the elasticity tensor; ε denotes the total strain tensor; ε_m and ε_t denotes the mechanical strain tensor and the thermal strain tensor, respectively. The thermal strain tensor ε_t can be calculated as (Zhou et al. 2022)

$$\varepsilon_t = \beta(T - \bar{T})I, \quad (7)$$

here β [K⁻¹] denotes the coefficient of thermal expansion of the rock matrix; $\bar{T}$ [K] represents the temperature.

The PF-CZM is employed to assess rock damage during LN₂ fracturing, and the associated damage evolution can be determined using the following damage-displacement coupled equations (Chen and Wu 2022)

$$\nabla \cdot \frac{2b}{c_a} G_f \nabla d - \frac{\partial \omega}{\partial d} H - \frac{\partial \alpha}{\partial d} \frac{G_f}{c_a b} = 0, \quad (8)$$

here b denotes the intrinsic length scale of the phase field; G_f [N/m] is the critical energy release rate; d is a phase-field parameter ($d(x, t) \in [0, 1]$, $d = 1$ and $d = 0$ represent the rock matrix is cracked and uncracked, respectively). The scaling constant c_a is set to π in this study. ω ($\omega(d) \in [0, 1]$) and α ($\alpha(d) \in [0, 1]$) represent the geometric crack function and the energetic degradation function, respectively. H [MPa] denotes the maximum effective crack driving force, which can be expressed as (Wu and Nguyen 2018, Wu and Chen 2021)

$$H = \max \left(\bar{Y}_0, \max_{n \in [0, T]} \bar{Y}_n \right), \quad (9)$$

where $\bar{Y}_n$ [MPa] represents the effective crack driving force; $\bar{Y}_0$ [MPa] represents the initial threshold crack driving force, which can be expressed as

$$\bar{Y} = \frac{\bar{\sigma}_1^2}{2\bar{E}_0}, \quad (10)$$

$$\bar{Y}_0 = \frac{f_t^2}{2\bar{E}_0}, \quad (11)$$

where $\bar{\sigma}_1$ [MPa] is the first stress; $\bar{E}_0 = \lambda_0 + 2\mu_0$ [MPa] is the elongation modulus. Here $\lambda_0 = \nu_0 E_0 / [(1 - 2\nu_0)(1 + \nu_0)]$ and $\mu_0 = E_0 / 2(1 + \nu_0)$ are the Lamé constants, where E_0 [MPa] and ν_0 denote the Young's modulus and Poisson's ratio, respectively, while f_t [MPa] represents the tensile strength.

Besides, the relationship between hydraulic parameters and phase-field parameter d can be written as

$$\begin{cases} \alpha_B = \alpha_r \chi_r + \alpha_f \chi_f \\ \varphi = \varphi_r \chi_r + \varphi_f \chi_f \\ k = k_r \chi_r + k_f \chi_f \\ \lambda = \lambda_r \chi_r + \lambda_f \chi_f \end{cases}, \quad (12)$$

where α_r , φ_r , k_r [m²] and λ_r [W/(m·K)] denote the Biot coefficient, porosity, permeability and thermal conductivity within the reservoir domain, respectively. Similarly, α_f , φ_f , k_f [m²] and λ_f [W/(m·K)] correspond to the same properties in the fracture domain.

The heterogeneity of the rock matrix is incorporated using a Weibull distribution (Wu 2021). Liquid nitrogen (LN₂) is employed as the fracturing fluid, saturating both the rock matrix pores and fractures.

Variations in the thermodynamic properties of nitrogen with temperature and pressure are characterized based on data from the NIST Chemistry WebBook (National Institute of Standards and Technology, USA).

Model solution

The workflow for geometric construction, assignment of initial and boundary conditions, and mesh generation is illustrated in Fig. 1. A 2D model is typically adopted, as accurate 3D simulations demand larger computational resources and longer solution times (Guo et al. 2020). To further reduce computational cost, only one-quarter of the rock domain is simulated, leveraging the symmetry inherent in the 2D model (Li et al. 2021). Consistent with our previous experimental studies (Hong et al. 2023), the rock length and borehole diameter are set to 0.1 m and 0.016 m, respectively. Low-temperature LN₂ is injected through the borehole, inducing fracture formation in the rock matrix and allowing fluid outflow at the outer boundary.

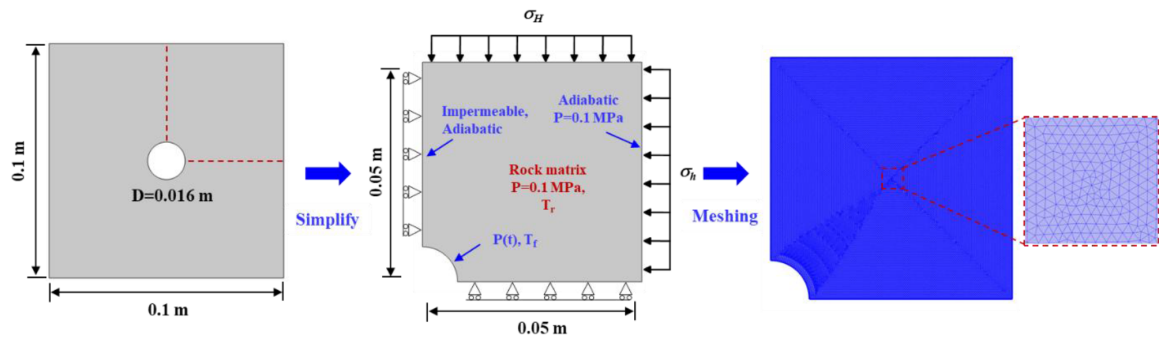


Figure 1—Workflow for geometric construction, assignment of initial and boundary conditions, and mesh generation.

Based on geological data from the FORGE (Frontier Observatory for Research in Geothermal Energy) site in the USA (EGI 2018) and well GR1 in the Gonghe Basin, China (Lei et al. 2019), the horizontal stress difference ratio and initial rock temperature are set to 0.4 ($\sigma_H/\sigma_h = 15/11$ MPa) and 200 °C, respectively. According to our prior experiments, the LN₂ injection temperature is −170 °C. At the outer boundary, a fixed initial pressure ($P = 0.1$ MPa), in-situ stress ($\sigma_H/\sigma_h = 15/11$ MPa) and no-heat transfer conditions are imposed. Symmetry boundaries are treated as adiabatic, with displacement constraints and no-flow conditions. The physical and mechanical properties of HDR rock for the 2D model are listed in Table 1 (Zhou and Zhuang 2020, Hong et al. 2023).

Table 1—Physical and mechanical properties of HDR rock.

Materials domain	Parameters	Value
Rock	Density [kg/m ³]	2650
	Specific heat capacity [J/(kg·K)]	1000
	Thermal expansion coefficient [K ⁻¹]	6×10 ⁻⁶
	Poisson's ratio	0.28
	Tensile strength [MPa]	10.0
	Fracture energy release rate [N/m]	60.0
	Young's modulus [GPa]	40.0
	Heterogeneity index	10
Reservoir domain	Permeability [m ²]	1.2×10 ⁻¹⁷
	Porosity	0.006
	Biot coefficient	0.5
	Thermal conductivity [W/(m·K)]	3.0

Materials domain	Parameters	Value
Fracture domain	Permeability [m^2]	1.0×10^{-10}
	Porosity	1.0
	Biot coefficient	1.0
	Thermal conductivity [$\text{W}/(\text{m} \cdot \text{K})$]	60.0

The computational domain is discretized using three-node triangular elements, with the mesh size and phase-field length scale b set to 0.3 mm and 0.9 mm, respectively, to ensure stable, smooth, and uniform fracture patterns. The 2D model consists of 68,924 elements and is solved in COMSOL Multiphysics using the finite element method. Critical variables including fluid pressure P , temperature T , displacement u and crack phase-field d are computed using a staggered solution scheme to enhance convergence efficiency. Due to the strong nonlinearity arising from coupled solid-fluid-heat interactions and complex fracture network formation, automatic Newton–Raphson iterations and Anderson acceleration techniques are applied. The absolute tolerance and maximum time step are set to 0.001 and 0.03 s, respectively. Model validation is presented in subsequent works (Yang et al. 2024, Gong et al. 2025).

Results

Fig. 2 illustrates three LN_2 injection schemes: balanced injection, pulsating injection, and stepped injection. The balanced injection method involves a constant injection pressure and a single injection of LN_2 . The pulsating pumping scheme entails cyclic injection and stoppage of low-temperature LN_2 , with an injection time of t_c , a stop injection time of t_s , and one complete cycle of "injection-stop". Similarly, the stepped pumping scheme involves periodic injection of LN_2 in a cyclic manner, with the injection pressure increasing step by step. In the base case, the LN_2 injection is conducted at a wellbore temperature of -170°C , with an injection time of 10 s and an injection pressure of 2 MPa. When the injection is halted, the wellbore temperature reverts to 200°C , representing the initial temperature of the formation rock. The stop injection time is set at 100 s, and the injection pressure is 0 MPa.

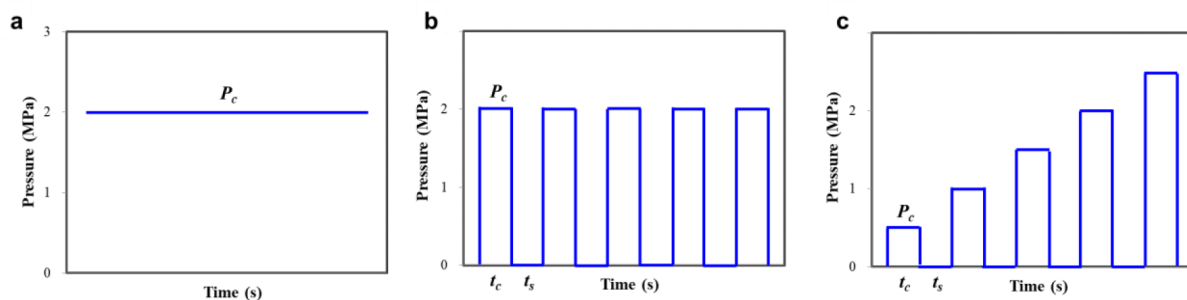


Figure 2— LN_2 injection schemes. (a) balanced injection; (b) pulsating injection; (c) stepped injection.

To compare the fracturing performance of balanced and pulsating injection schemes with equal LN_2 injection time, the total injection time is set at 50 s while maintaining a constant injection pressure of 2 MPa. In the balanced injection scheme, LN_2 is continuously injected at 2 MPa for 50 s. On the other hand, the pulsating injection scheme involves injecting LN_2 at 2 MPa for 10 s, followed by a 100 s pause, with a cumulative total of 5 cycles ($C_n=5$). Fig. 3 depicts the comparison of fracture patterns between balanced injection and pulsating injection, while Fig. 4 illustrates the comparison of damage ratios. It is observed that when the cycle number is 1, the LN_2 injection time is 10 s, identical to that of the balanced injection cycle, resulting in consistent fracture patterns and damage ratios. However, with an increase in the cycle number

to 2, corresponding to a cumulative LN₂ injection time of 20 s, two groups of significant fractures propagate along the direction of the minimum horizontal stress in the pulsating injection scheme, whereas the fractures continue to extend forward. In contrast, balanced injection does not generate new fractures and only extends fractures along the direction of the minimum horizontal stress. Additionally, the damage ratio of pulsating injection is 34.4% higher than that of balanced injection. As the cycle number of pulsating injections reaches 3, with a corresponding cumulative LN₂ injection time of 30 s, induced fractures continue to extend, causing further damage. Nonetheless, the damage ratio of pulsating injection remains 15.1% higher compared to balanced injection. For cycle numbers 4 and 5 in pulsating injection schemes, the corresponding cumulative LN₂ injection times are 40 s and 50 s. In both balanced and pulsating injection schemes, no new fractures are generated. Moreover, in the pulsating injection scheme, the damage remains relatively unchanged (Fig. 4a). However, in the balanced injection scheme, fractures continue to extend, leading to an increase in the damage proportion, although at a slower rate. At a cumulative injection time of 50 s, the damage ratio of balanced injection is even 6.1% higher than that of pulsating injection, with a larger fracture extension range.

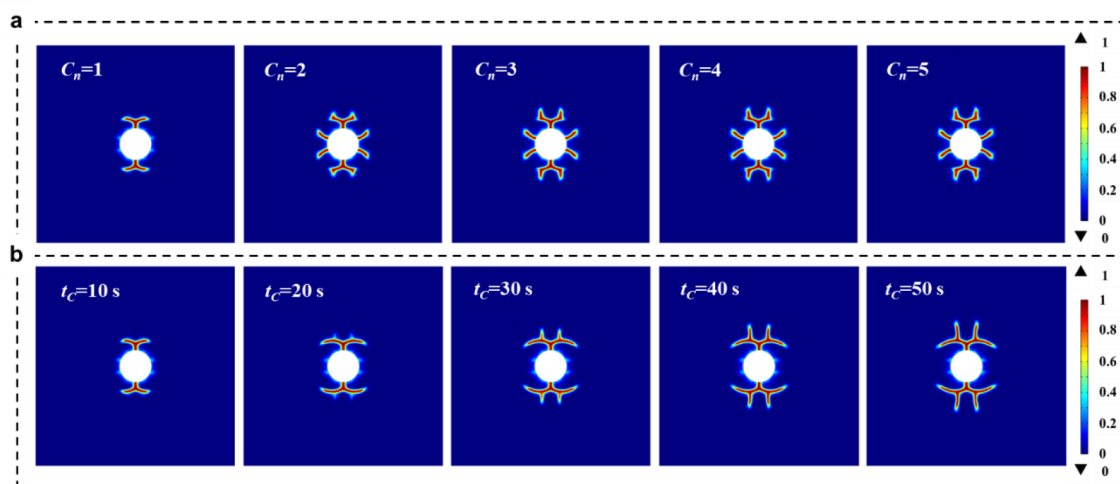


Figure 3—Comparison of fracture patterns under different LN₂ injection schemes. (a) pulsating injection; (b) balanced injection.

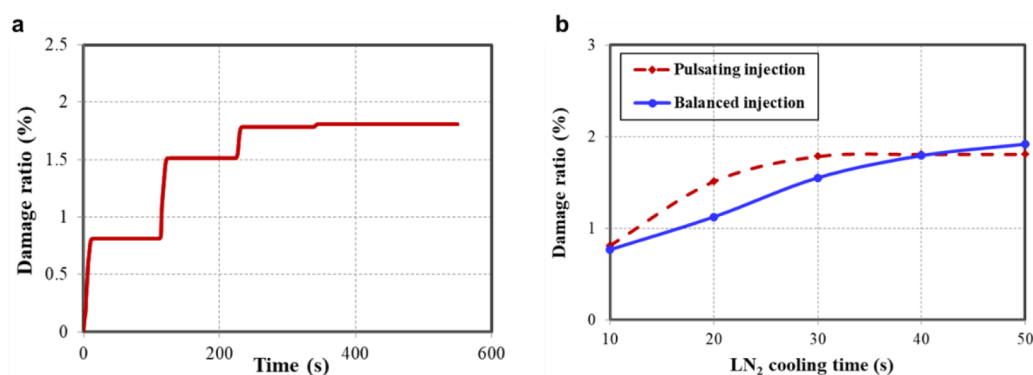


Figure 4—Comparison of damage ratios under different LN₂ injection schemes. (a) damage ratio changes with time under pulsating injection schemes; (b) pulsating injection and balanced injection.

In summary, cyclic thermal interactions between LN₂ and high-temperature rock can induce significant thermal fatigue damage, thereby enhancing overall fracturing performance. During cyclic LN₂ injection, the coupled thermal stresses and fluid pressures periodically intensify and relax, resulting in the repeated opening and closing of fractures. Exposure to freeze-thaw cycles, stress oscillations, and fatigue can

further promote the development of more intricate fracture networks and potentially increase the stimulated reservoir volume. Compared with the balanced injection scheme, the pulsating injection scheme exhibits a greater potential for generating complex fracture geometries. However, after 3 cycles, the fractures around the wellbore become complicated, and it becomes difficult to generate new fractures due to the stress shadow effect. Additionally, there is a stop injection time in the pulsating injection scheme, and the low-temperature sweep range of LN₂ is small within the total injection time equivalent to that of the balanced injection scheme. Therefore, the fracture patterns and damage ratios of LN₂ injection are not significantly improved after 3 cycles in the pulsating injection scheme. Hence, a combination of the pulsating and balanced injection schemes can be adopted. In the early stage, the pulsating injection can improve the fracturing performance around the wellbore. In the later stage, LN₂ can be continuously injected with a pulsating injection scheme to improve the low temperature expansion range and promote the fracture to extend further to the depth of the formation.

The impact of injection pressure on fracture patterns and damage ratios is further examined in the pulsating injection scheme, with an injection time of 10 s, a stop time of 100 s, and a cycle number of 3. Fig. 5 illustrates that, at injection pressures of 1, 2, and 3 MPa, thermal stress dominates, leading to the formation of complex fractures around the well, with minimal change in the damage ratio. In contrast, at 4 and 6 MPa, only biplane bifurcation fractures appear, while at 10 MPa, fluid pressure becomes the primary factor, resulting in a single main fracture with greater extension. Notably, the increase in injection pressure from 1 MPa to 10 MPa causes a 25.0% decrease in the damage ratio, indicating that higher pressure promotes fracture extension but hinders the formation of complex fractures around the well. It is clear that a balanced distribution of thermal stress and fluid pressure is essential for promoting fracture bifurcation and the development of a complex fracture network during cyclic LN₂ fracturing in HDR reservoirs. Excessive fluid pressure tends to favor the formation of a single dominant fracture, whereas overly high thermal stress primarily induces thermally-driven fractures near the wellbore, which can hinder the branching and redirection of the main fracture.

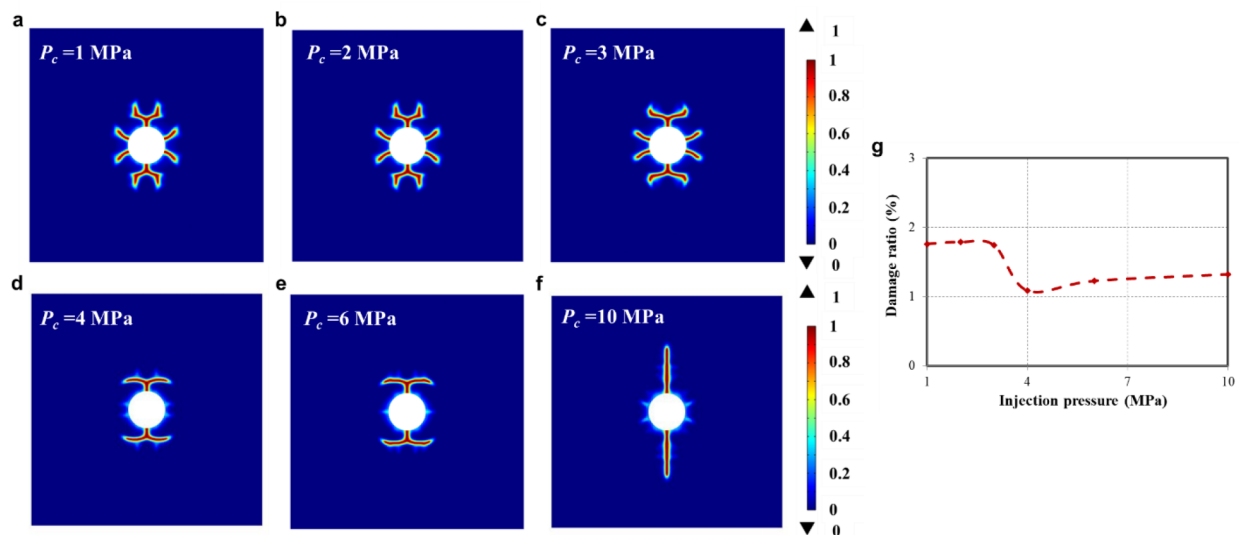


Figure 5—Effect of the injection pressure (P_c) on fracture patterns and damage ratios.
(a) 1 MPa; (b) 2 MPa; (c) 3 MPa; (d) 4 MPa; (e) 6 MPa; (f) 10 MPa; (g) damage ratio

In this section, four stepped injection schemes are analyzed, with an injection time of 10 s, injection stop time of 100 s, and cycle number of 5. The injection pressure for each case increases progressively, with values set to 1/2/3/4/5 MPa, 2/4/6/8/10 MPa, 3/6/9/12/15 MPa, and 4/8/12/16/20 MPa in each cycle, respectively. Fig. 6 demonstrates that when adopting Case 1, where the injection pressure in each cycle is

low, thermal stress governs, leading to the formation of a complex fracture network around the wellbore, albeit with limited main fracture extension. In contrast, with Cases 2 and 3, the steadily rising injection pressure shifts the dominance to fluid pressure, prompting continuous fracture extension towards the direction of minimum horizontal stress, while notably reducing the damage proportion. In case 4, the substantial pressure increase leads to significant fracture extension along the the direction of minimum horizontal stress. Eventually expanding towards the direction of maximum horizontal stress under in-situ stress control, resulting in a 60.8% increase in the damage ratio compared to case 1, with more tortuous and prolonged fractures. Accordingly, the stepped injection scheme with higher injection pressure is expected to induce the major fracture away from the direction of minimum horizontal stress in the near-wellbore area.

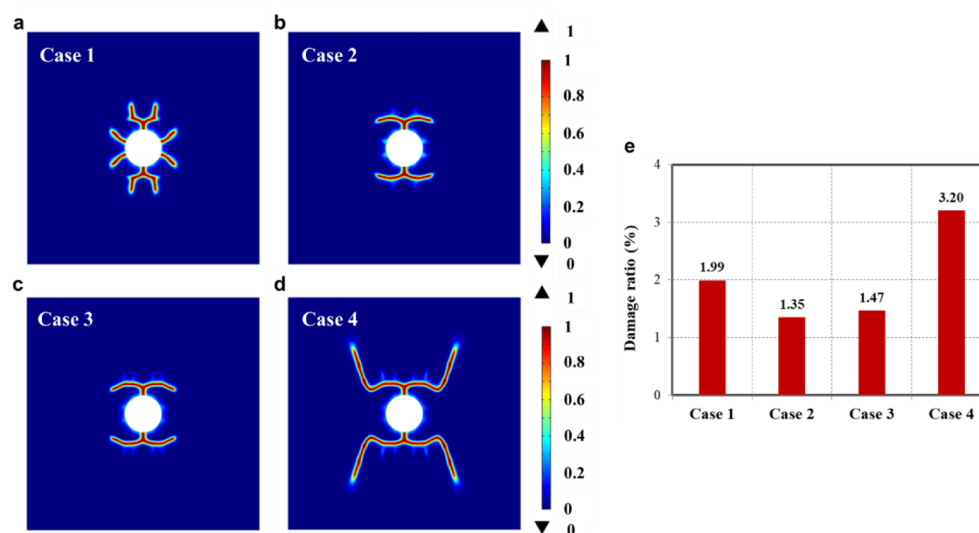


Figure 6—Comparison of fracture patterns and damage ratios under different stepped injection schemes. (a) case 1; (b) case 2; (c) case 3; (d) case 4; (e) damage ratio.

Conclusions and Recommendations

This study aims to assess the feasibility of cyclic LN₂ fracturing in HDR reservoirs. A thermo-hydro-mechanical-damage (THMD) coupling framework, incorporating rock heterogeneity and based on the phase-field cohesive zone method (PF-CZM), is developed to investigate fracture initiation, propagation, and the formation of complex fracture networks during cyclic LN₂ injection. The fracturing performance of three injection strategies—balanced, pulsating, and stepped schemes—was systematically evaluated. The key findings are summarized as follows.

In the pulsating injection scheme, the damage ratio increased by 120.1% when the cycle number increased from 1 to 3. Under a fixed LN₂ injection time, pulsating injection demonstrates superior effectiveness compared to balanced injection, but the increase in damage becomes less pronounced after 3 cycles. The results indicate that **alternating thermal interactions** between LN₂ and high-temperature rock during cyclic injection can induce **enhanced thermal fatigue damage**, thereby improving overall fracturing performance.

A balanced distribution of thermal stress and fluid pressure is essential for promoting fracture bifurcation and the development of a complex fracture network during cyclic LN₂ fracturing in HDR reservoirs. When fluid pressure is excessively high, it tends to produce a single dominant fracture, whereas overly strong thermal stress primarily induces thermally driven fractures near the wellbore, which can obstruct the branching and redirection of the main fracture.

In the stepped injection scheme, a relatively low injection pressure promotes the development of numerous thermally induced fractures around the wellbore. However, the primary fracture exhibits limited extension. In contrast, a higher injection pressure tends to drive the main fracture to propagate farther from the direction of minimum horizontal stress in the near-wellbore region.

Additional research is required to optimize injection schemes. Besides, field tests are recommended to further verify the advantages of cyclic LN₂ fracturing in exploiting deep geothermal energy resources, as all results were obtained through numerical investigation.

Nomenclature

EGS	enhanced geothermal system
HDR	hot dry rock
THMD	thermo-hydro-mechanical-damage
LN ₂	liquid nitrogen
CSS	cyclic soft stimulation
FHF	fatigue hydraulic fracturing
PF-CZM	phase-field cohesive zone method
NIST	National Institute of Standards and Technology
FORGE	Frontier Observatory for Research in Geothermal Energy
ρ_f	fracturing fluid density
α_B	Biot coefficient
$\mathbf{v}$	Darcy's velocity
q_m	fluid source term
P	fluid pressure
χ_r	indicator function
ε_{vol}	volumetric strain
S	storage coefficient
ϕ	porosity of the rock matrix
c_f	fluid compressibility
K_V	bulk modulus
$(\rho C_p)_{eff}$	effective specific heat capacity
λ_{eff}	effective thermal conductivity
$C_{p,f}$	specific heat capacity of fracturing fluid
T	temperature in the current state
Q_T	heat source term
$\mathbf{b}$	body force
$\boldsymbol{\sigma}$	Cauchy stress tensor
$\boldsymbol{\sigma}_{eff}$	effective stress tensor
$\mathbf{I}$	identity tensor
$\boldsymbol{\sigma}_{eff}$	average stress
$\mathbf{C}$	elasticity tensor
$\boldsymbol{\varepsilon}$	total strain tensor
$\boldsymbol{\varepsilon}_m$	mechanical strain tensor
$\boldsymbol{\varepsilon}_t$	thermal strain tensor
β	thermal expansion coefficient
b	phase-field intrinsic length scale
G_f	critical energy release rate
d	phase-field parameter

- c_a scaling constant
- ω geometric crack function
- α energetic degradation function
- H the maximum value of effective crack driving force
- Y the effective crack driving force
- λ_0 Lamé constants of isotropic elasticity
- μ_0 Lamé constants of isotropic elasticity
- f_t tensile strength of the rock matrix

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